

# Visual Perception for Robot Loco- & Manipulation



**Invited Speaker**

**Tianwei Zhang**

*Shenzhen Institute of Artificial  
Intelligence and Robotics for Society*

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**Time:** 14:00 (HKT)

**Zoom Meeting:** 801 137 0362

## Biography

Tianwei Zhang is an Associate Research Scientist at Shenzhen Institute of Artificial Intelligence and Robotics for Society, Research Assistant Professor in the Chinese University of Hongkong, Shenzhen. He received Doctor Degree from the Department of Mechano-Informatics, The University of Tokyo, 2019. His research interests are visual manipulation and dynamic SLAM. He has published several articles in ICRA/IROS/RAL.

## Abstract

Robot perception includes many kinds of sensory data processing and artificial intelligence/machine learning (AI/ML) techniques. Examples of robot perception capabilities include object detection, environment understanding, state estimation, and human/pedestrian detection, among others. It is related to many applications in robotics such as path planning, control and loco-manipulation. due to progress in the use of robots in unstructured environments, robots are increasingly being used in non-traditional application domains ranging from collaborative manufacturing to assisted living environments. In these shared autonomy scenarios, robot perception system needs the ability to efficiently and accurately model and predict human motions to improve safety and efficiency in Human Robot Collaboration (HRC) scenarios.